

Zekun Song

Toronto, ON | +1 (647) 829-6198 | zekun.song@mail.utoronto.ca
linkedin.com/in/zekun-song | github.com/UTSGJohnsonSong

Backend & Data Engineer | Clinical AI | Reliable Data Platforms

GPA: 3.87/4.00 (2025-26) | 8-month co-op: Sep 2026 - Apr 2027 | Canadian permanent resident - no sponsorship required

EDUCATION

University of Toronto (St. George)
Honours Bachelor of Science - Double Specialist in Computer Science and Data Science
Honours: Dean's List Scholar; Reuben Wells Leonard Scholarship
Coursework: Databases, Software Engineering, Machine Learning, Data Science I & II, Statistical Modelling, Time Series Analysis

Sep 2023 - Dec 2027 (Expected)
Toronto, ON

EXPERIENCE

Remeda - Hospital-Partnered Clinical AI Startup
Backend & Data Engineer | Clinical AI Platform (CSC494 Venture)

Apr 2026 - Present
Toronto, ON

- Clinical Data:** Owned the terminology layer of an OMOP warehouse integrating FHIR, NLP, Synthea, and MIMIC-IV data.
- Terminology ETL:** Integrated 12 vocabularies and 90,000+ codes through APIs, bulk loads, and Python ETL pipelines.
- Normalization API:** Built a FastAPI/asyncpg service mapping clinical text to SNOMED CT, ICD-10, RxNorm and LOINC.
- Data Reliability:** Made ETL rerun-safe with batched SQL, SHA-256-pinned reference data, and 60+ unit tests.
- Data Validation:** Prevented an underdetermined metric from shipping after checking values against CMS specifications.
- Production Debugging:** Root-caused a demo-blocking stale backend address and led end-to-end validation.

Shell (Shanghai) Technology Co., Ltd.
Data Science Intern

Jun 2024 - Sep 2024
Shanghai, China

- Self-Service Analytics:** Delivered an internal portal that cut weekly ad-hoc data requests from 16 to 7.
- NL-to-SQL Validation:** Integrated LLM-to-SQL with schema and permission checks; 92% pass across 300+ tests.
- Data Automation:** Designed query logs and user analytics, cutting the product-feedback cycle from three weeks to one.
- BI Reporting:** Built 10+ auto-refreshed Power BI dashboards over SQL pipelines for internal KPI reporting.
- Stakeholder Delivery:** Translated business requirements into API contracts and a high-fidelity Figma prototype.

WholeViz - Industry Partner Capstone
Technical Product Lead | Full-Stack Analytics Platform

Sep 2025 - Dec 2025
Toronto, ON

- Team Leadership:** Led a 6-person team to ship a Django/PostgreSQL analytics platform for non-technical users.
- Backend & Data:** Designed normalized schemas, multi-user access, and ingestion, validation, and storage pipelines.
- Product Design:** Defined self-service data exploration workflows and usability-driven features for non-technical users.
- Partner Delivery:** Turned stakeholder needs into specifications through agile iterations and industry-partner handoff.

PROJECTS

CardioTrace | *Python, PostgreSQL, dbt, Survival Analysis*

Jun 2026 - Present

- Cohort Engineering:** Built a reproducible NHANES/mortality pipeline spanning 11 cycles and 235,553 person-years.
- Survival Modelling:** Trained survey-weighted competing-risk Cox models on 20,736 adults and 925 CVD deaths.
- Methodology:** Applied survey weights, competing-risk methods, and a forward-time split to prevent cohort leakage.
- Temporal Validation:** Achieved 10-year C-index 0.804; predicted risk was within 0.06 pp of observed.
- Data Integrity:** Found CDC renames dropping lab data for 15,332 people; withdrew results and rebuilt 64-test ingestion.

Personal Job Application OS | *FastAPI, Next.js, SQLAlchemy*

Jul 2026 - Present

- Platform Engineering:** Built a FastAPI/Next.js platform normalizing 10,921 postings from 58 first-party ATS sources.
- Decision Engine:** Implemented deduplication, eligibility gates, role classification, and evidence-weighted ranking.
- Data Model:** Designed a 63-table SQLAlchemy/Alembic schema and append-only audit trail.
- Submission Safety:** Enforced hash-bound per-application confirmation; 280 pytest cases pass.

TECHNICAL SKILLS

Languages Python, SQL, TypeScript/JavaScript, R, Java, Bash	Backend FastAPI, Django, REST APIs, asyncpg, SQLAlchemy/Alembic	Data & AI PostgreSQL, dbt, Pandas, NumPy, Scikit-learn, XGBoost, SHAP	Delivery & BI Git/GitHub, Docker, pytest, GitHub Actions, Linux, Power BI, Vercel, Render
---	---	---	---